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CMPUT 365: Introduction to Reinforcement Learning

Worksheet 8: Planning, Learning & Acting

Question 1. An agent observes the following two episodes from an MDP,

$$S_0 = 0, A_0 = 1, R_1 = 1, S_1 = 1, A_1 = 1, R_2 = 1$$

$$S_0 = 0, A_0 = 0, R_1 = 0, S_1 = 0, A_1 = 1, R_2 = 1, S_2 = 1, A_2 = 1, R_3 = 1$$

and updates its deterministic model accordingly. What would the model output for the following queries:

(a) Model($S = 0, A = 0$):

(b) Model($S = 0, A = 1$):

(c) Model($S = 1, A = 0$):

(d) Model($S = 1, A = 1$):

Question 2. An agent is in a 4-state MDP, $\mathcal{S} = \{1, 2, 3, 4\}$, where each state has two actions $\mathcal{A} = \{1, 2\}$. Assume the agent saw the following trajectory,

$$\begin{aligned}
&S_0 = 1, A_0 = 2, R_1 = -1,\\
&S_1 = 1, A_1 = 1, R_2 = 1,\\
&S_2 = 2, A_2 = 2, R_3 = -1,\\
&S_3 = 2, A_3 = 1, R_4 = 1,\\
&S_4 = 3, A_4 = 1, R_5 = 100,\\
&S_5 = 4
\end{aligned}$$

and uses Tabular Dyna-Q with 5 planning steps for each interaction with the environment.

(a) Once the agent sees S_5, how many Q-learning updates has it done with **real experience**? How many updates has it done with **simulated experience**?

(b) Which of the following are possible (or not possible) simulated transitions $\{S, A, R, S'\}$ given the above observed trajectory with a deterministic model and random search control?

(i) $\{S = 1, A = 1, R = 1, S' = 2\}$

(ii) $\{S = 2, A = 1, R = -1, S' = 3\}$

(iii) $\{S = 2, A = 2, R = -1, S' = 2\}$

(iv) $\{S = 1, A = 2, R = -1, S' = 1\}$

(v) $\{S = 3, A = 1, R = 100, S' = 5\}$

Question 3. Modify the Tabular Dyna-Q algorithm so that it uses Expected Sarsa instead of Q-learning. Assume that the target policy is ϵ-greedy. What should we call this algorithm?

Tabular Dyna-Q

Initialize $Q(s,a)$ and $Model(s,a)$ for all $s \in \mathcal{S}$ and $a \in \mathcal{A}(s)$
Loop forever:
(a) $S \leftarrow$ current (nonterminal) state
(b) $A \leftarrow \varepsilon$-greedy(S,Q)
(c) Take action A; observe resultant reward, R, and state, S'
(d) $Q(S,A) \leftarrow Q(S,A) + \alpha\big[R + \gamma \max_a Q(S',a) - Q(S,A)\big]$
(e) $Model(S,A) \leftarrow R, S'$ (assuming deterministic environment)
(f) Loop repeat n times:
$S \leftarrow$ random previously observed state
$A \leftarrow$ random action previously taken in S
$R, S' \leftarrow Model(S,A)$
$Q(S,A) \leftarrow Q(S,A) + \alpha\big[R + \gamma \max_a Q(S',a) - Q(S,A)\big]$

Question 4. Consider an MDP with two states $\{1,2\}$ and two possible actions: $\{\text{stay}, \text{switch}\}$. The state transitions are deterministic, the state does not change if the action is "stay" and the state switches if the action is "switch". However, rewards are randomly distributed:

$$P(R=r|S=1,A=\text{stay}) = \begin{cases} 0.4 & \text{for } r=0 \\ 0.6 & \text{for } r=1 \end{cases}, \quad P(R=r|S=1,A=\text{switch}) = \begin{cases} 0.5 & \text{for } r=0 \\ 0.5 & \text{for } r=1 \end{cases}$$

$$P(R=r|S=2,A=\text{stay}) = \begin{cases} 0.6 & \text{for } r=0 \\ 0.4 & \text{for } r=1 \end{cases}, \quad P(R=r|S=2,A=\text{switch}) = \begin{cases} 0.5 & \text{for } r=0 \\ 0.5 & \text{for } r=1 \end{cases}$$

(a) How might you learn the reward model? Hint: think about how probabilities are estimated. For example, what if you were to estimate the probability of a coin landing on heads? If you observed 10 coin flips with 8 heads and 2 tails, then you can estimate the probabilities by counting: $p(\text{heads}) = \frac{8}{10} = 0.8$ and $p(\text{tails}) = \frac{2}{10} = 0.2$.
(b) Modify the tabular Dyna-Q algorithm to handle this MDP with stochastic rewards.

Question 5. Consider an MDP with three states $\mathcal{S} = \{1,2,3\}$, where each state has two possible actions $\mathcal{A} = \{1,2\}$ and a discount rate $\gamma = 0.5$. Suppose estimates of $Q(S,A)$ are initialized to 0 and you observed the following episode according to an unknown behaviour policy where S_3 is the terminal state.

$$S_0 = 1, A_0 = 1, R_1 = -7, S_1 = 2, A_1 = 2, R_2 = 5, S_2 = 1, A_2 = 1, R_3 = 10$$

(a) Suppose you used Q-learning with the above trajectory to estimate $Q(S,A)$, what are your new estimates for $Q(S=1,A=1)$ using $\alpha = 0.1$?
(b) What is one possible model for this environment? Is the model stochastic or deterministic?
(c) Suppose in the planning loop, after search control, we would like to update $Q(S=1,A=1)$ with Q-planning. What are the possible outputs of Model$(S=1,A=1)$?
(d) If your model outputs $R = R_3$ and $S' = S_3$, what is $Q(S=1,A=1)$ after one Q-planning update? Use the estimates of $Q(S,A)$ from before.

Question 6. Exercise 8.2 from Sutton & Barto (2018).

Question 7. Exercise 8.3 from Sutton & Barto (2018).